Assignment 3

Question 1

hostname	ipconfig	getmac	ping	arp
nbstate	route	path	pathping	netstat
tracert				

Practice the commands mentioned in the above table and fill in the blanks.

Solution

Command	Working
ping	Command is one of the most often used networking utilities for detecting devices on a network and for troubleshooting network problems.
hostname	command that displays the hostname of your machine
ipconfig	frequently used utility that is used for finding network information about your local machine like ip addresses, dns addresses etc.
nbstat	diagnostic tool for troubleshooting netbios problems
netstat	used for displaying information about tcp and udp connections and ports.
getmac	command that shows the mac address of your network interfaces
arp	for showing the address resolution cache. this command must be used with a command line switch -a is the most common.
tracert	command prints the path. if all routers on the path are functional, this command prints the full path.
path	command specifies the location where ms-dos should look when it executes a command.
route	command allows you to make manual entries into the network routing tables.
pathping	after sending out packets from you to a given destination, it analyzes the route taken and computes packet loss on a per-hop basis.

Outputs

`ipconfig`

txt

```txt
Windows IP Configuration


Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::5ca6:b97c:1b7f:ce94%9
   IPv4 Address. . . . . . . . . . . : 172.16.119.58
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
```

```
   Default Gateway . . . . . . . . . : 172.16.119.1

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 6:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

ping

txt

```
Pinging google.com [172.217.26.14] with 32 bytes of data:
Reply from 172.217.26.14: bytes=32 time=10ms TTL=117
Reply from 172.217.26.14: bytes=32 time=10ms TTL=117

Ping statistics for 172.217.26.14:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
```

hostname

txt

```
DESKTOP-IDSI54N
```

netstat

txt

```
Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:49674        DESKTOP-IDSI54N:49675  ESTABLISHED
  TCP    127.0.0.1:49675        DESKTOP-IDSI54N:49674  ESTABLISHED
  TCP    127.0.0.1:59005        DESKTOP-IDSI54N:59436  ESTABLISHED
  TCP    127.0.0.1:59006        DESKTOP-IDSI54N:59008  ESTABLISHED
  TCP    127.0.0.1:59008        DESKTOP-IDSI54N:59006  ESTABLISHED
  TCP    127.0.0.1:59333        DESKTOP-IDSI54N:59005  TIME_WAIT
  TCP    127.0.0.1:59372        DESKTOP-IDSI54N:59005  TIME_WAIT
  TCP    127.0.0.1:59394        DESKTOP-IDSI54N:59005  TIME_WAIT
  TCP    127.0.0.1:59419        DESKTOP-IDSI54N:59005  TIME_WAIT
  TCP    127.0.0.1:59436        DESKTOP-IDSI54N:59005  ESTABLISHED
  TCP    172.16.119.58:58959    a23-208-178-166:https  ESTABLISHED
```

getmac

txt

```
Physical Address    Transport Name
=================== ==========================================================
D8-9C-67-6F-4E-1F   Media disconnected
F4-39-09-22-9E-A7   \Device\Tcpip_{65BECBCF-235F-424E-82A7-FE2902C3E542}

arp -a
Interface: 172.16.119.58 --- 0x9
```

```txt
  Internet Address      Physical Address      Type
  172.16.119.1          74-8e-f8-90-80-78     dynamic
  172.16.119.21         f4-39-09-22-9e-68     dynamic
  172.16.119.27         f4-39-09-22-9d-b8     dynamic
  172.16.119.54         f4-39-09-22-9f-2a     dynamic
  172.16.119.88         f4-39-09-22-9d-c9     dynamic
  172.16.119.89         f4-39-09-22-9f-15     dynamic
  172.16.119.90         f4-39-09-22-9e-08     dynamic
  172.16.119.91         f4-39-09-22-9d-ac     dynamic
  172.16.119.95         f4-39-09-22-9d-8e     dynamic
  172.16.119.96         f4-39-09-22-9e-73     dynamic
  172.16.119.101        f4-39-09-22-9d-ab     dynamic
  172.16.119.102        f4-39-09-22-9d-cf     dynamic
  172.16.119.105        f4-39-09-22-9d-d9     dynamic
  172.16.119.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.250.250       01-00-5e-7f-fa-fa     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
```

arp -a

```txt
Interface: 172.16.119.58 --- 0x9
  Internet Address      Physical Address      Type
  172.16.119.1          74-8e-f8-90-80-78     dynamic
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  172.16.119.54         f4-39-09-22-9f-2a     dynamic
  172.16.119.88         f4-39-09-22-9d-c9     dynamic
  172.16.119.89         f4-39-09-22-9f-15     dynamic
  172.16.119.90         f4-39-09-22-9e-08     dynamic
  172.16.119.91         f4-39-09-22-9d-ac     dynamic
  172.16.119.95         f4-39-09-22-9d-8e     dynamic
  172.16.119.96         f4-39-09-22-9e-73     dynamic
  172.16.119.101        f4-39-09-22-9d-ab     dynamic
  172.16.119.102        f4-39-09-22-9d-cf     dynamic
  172.16.119.105        f4-39-09-22-9d-d9     dynamic
  172.16.119.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.250.250       01-00-5e-7f-fa-fa     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
```

tracert -h 5 google.com

```txt
Tracing route to google.com [142.251.221.238]
over a maximum of 5 hops:

  1    <1 ms     2 ms     1 ms  172.16.119.1
  2    <1 ms    <1 ms    <1 ms  172.31.1.6
  3    20 ms    25 ms    36 ms  112.196.126.1
```

```txt
  4     5 ms     4 ms     4 ms  202.164.32.25
  5     4 ms     4 ms     4 ms  202.164.51.36

Trace complete.
```

path

```txt
PATH=C:\ProgramData\Oracle\Java\javapath;C:\Program Files (x86)\Intel\Intel(R) Management Engine Components\iCLS\;C:\Program Files\Intel\Intel(R) Management Engine Components\iCLS\;C:\WINDOWS\system32;C:\WINDOWS;C:\WINDOWS\System32\Wbem;C:\WINDOWS\System32\WindowsPowerShell\v1.0\;C:\WINDOWS\System32\OpenSSH\;C:\Program Files (x86)\Intel\Intel(R) Management Engine Components\DAL;C:\Program Files\Intel\Intel(R) Management Engine Components\DAL;C:\Program Files (x86)\Intel\Intel(R) Management Engine Components\IPT;C:\Program Files\Intel\Intel(R) Management Engine Components\IPT;C:\Program Files\dotnet\;C:\Users\CSED\AppData\Local\Microsoft\WindowsApps;
```

routeprint

```txt
===========================================================================
Interface List
  9...f4 39 09 22 9e a7 ......Intel(R) Ethernet Connection (7) I219-LM #2
 17...da 9c 67 6f 4e 1f ......Microsoft Wi-Fi Direct Virtual Adapter #5
 10...fa 9c 67 6f 4e 1f ......Microsoft Wi-Fi Direct Virtual Adapter #6
  1...........................Software Loopback Interface 1
 14...d8 9c 67 6f 4e 1f ......Realtek RTL8821CE 802.11ac PCIe Adapter #2
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0     172.16.119.1    172.16.119.58    281
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
     172.16.119.0    255.255.255.0         On-link     172.16.119.58    281
    172.16.119.58  255.255.255.255         On-link     172.16.119.58    281
   172.16.119.255  255.255.255.255         On-link     172.16.119.58    281
        224.0.0.0        240.0.0.0         On-link         127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link     172.16.119.58    281
  255.255.255.255  255.255.255.255         On-link         127.0.0.1    331
  255.255.255.255  255.255.255.255         On-link     172.16.119.58    281
===========================================================================
Persistent Routes:
  Network Address          Netmask  Gateway Address  Metric
          0.0.0.0          0.0.0.0     172.16.119.1  Default
===========================================================================

IPv6 Route Table
===========================================================================
Active Routes:
 If Metric Network Destination      Gateway
  1    331 ::1/128                  On-link
  9    281 fe80::/64                On-link
```

```
  9    281 fe80::5ca6:b97c:1b7f:ce94/128
                                    On-link
  1    331 ff00::/8                 On-link
  9    281 ff00::/8                 On-link
===========================================================================
Persistent Routes:
  None
```